

Do Synthetic Personas Predict Real Audience Response?

A Sim-to-Real Study Where a No-Persona Baseline Beats Persona-Based Copy Simulation

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Abstract

Marketers increasingly use large language models (LLMs) as “synthetic personas” to predict how an audience will react to a piece of copy before it ships, encouraged by evidence that profile-conditioned LLMs mimic human samples. But is that prediction actually valid against real behaviour—and does the persona machinery help? We present a sim-to-real validity study using the Upworthy Research Archive—thousands of headline A/B tests on shared real traffic, with measured click-through—as held-out ground truth. We compare a ten-persona panel, grounded in the real audience’s demographics, against a no-persona zero-shot baseline that simply asks the model how likely a typical reader is to click. Two findings stand out. First, **ground-truth reliability is the binding constraint**: most A/B tests have no statistically distinguishable winner, so validity can only be measured on the reliable subset ($n = 399$). Second, and counter to the persona-simulation premise, **persona conditioning degrades predictive validity**: the no-persona baseline ranks variants markedly better (Kendall $\tau = 0.361$, a medium effect; top-1 accuracy 49.2%) than the persona panel ($\tau = 0.084$; top-1 34.6%), with non-overlapping confidence intervals. Asking the model directly taps an accurate population-level prior; forcing it to role-play specific personas injects bias and noise. The result replicates across three independent Upworthy splits, holds in direction on a different-domain news dataset, and is robust to seed, prompt phrasing, and model choice—across three Gemini tiers and a different model family (OpenAI gpt-4.1, significant paired gap). The takeaway: for predicting aggregate engagement, a plain LLM ranker beats persona simulation—synthetic personas are not merely a weak predictor, they are worse than not using them. All numbers regenerate from a public, artifact-first replication package.

1 Introduction

Before launching a campaign, marketers want to know which version of a message will land. A fast-growing practice replaces (or precedes) live A/B testing with *synthetic audience simulation*: a large language model (LLM) is conditioned on a set of audience “personas” and asked to react to each candidate message, and the message its personas prefer is shipped. The appeal is obvious—instant, cheap feedback—and it is encouraged by findings that profile-conditioned LLMs can reproduce aspects of real human samples, an idea sometimes called “silicon sampling” [1].

The appeal, however, outruns the evidence. The premise that a synthetic persona predicts how a *real* audience behaves is rarely tested against real outcomes, because doing so requires ground truth that pairs concrete copy with measured audience response. We borrow the *sim-to-real* framing from robotics—how well does behaviour measured in simulation transfer to the real world?—and instantiate it with the Upworthy Research Archive [13]: thousands of A/B

tests (which we call *packages*) in which competing headline variants were shown to the same real traffic and their click-through rates recorded. This lets us ask, for held-out copy never seen during persona design, whether the simulator orders the variants the way the audience did.

We report two main findings. First, the dominant obstacle is not the model but the data: in most A/B tests the real winner is not statistically distinguishable from the runner-up, so no predictor can recover it—validity is only meaningful on the reliable subset. Second, and contrary to the premise of persona simulation, *persona conditioning hurts*: a no-persona zero-shot baseline that simply asks the model how likely a typical reader is to click ranks the variants markedly better (a medium effect) than the demographically grounded ten-persona panel (a small, barely-significant effect), with non-overlapping confidence intervals. The base model holds an accurate population-level prior on clickability; forcing it to role-play specific personas pulls it away from that prior. We pre-register effect-size thresholds and re-

port seed-stability and a model comparison so neither effect is overstated.

Contributions.

1. A reproducible *sim-to-real* validity protocol for copy simulation, using a public, held-out A/B-tested benchmark [13] as ground truth.
2. A ground-truth-reliability filtering step (significant-winner A/B tests) that prevents validating a predictor against statistical noise [11].
3. A within-package ranking evaluation with a construct-matched click-intent elicitation, a no-persona zero-shot baseline that isolates the value of persona conditioning, pre-registered effect-size thresholds [3], bootstrap CIs, seed-stability, and a cost/quality model comparison.
4. An artifact-first replication package: every reported number is regenerated from aggregated artifacts by public scripts.

Research questions.

RQ1 Do the personas’ predicted ranking match the real click-through ranking on copy with a reliable winner, and do they beat a no-persona zero-shot baseline?

RQ2 Does predictive validity increase with the strength of the real effect?

RQ3 Is the result robust to the intent-vs-engagement construct gap, to seed sampling, and to model choice?

2 Related Work

LLMs as proxies for human populations. A growing body of work uses language models conditioned on demographic or attitudinal profiles to approximate human samples—“silicon sampling” [1]—and surveys the promise and limits of LLMs as tools for computational social science [19]. The evidence, however, is mixed. On the supportive side, profile-conditioned models can reproduce aggregate survey *distributions* [1], and agents grounded in a two-hour interview *per person* replicate that individual’s survey responses at $\approx 85\%$ of the person’s own two-week test-retest reliability [14]—though both successes are on *stated attitudes*, and the latter buys its fidelity with rich per-person grounding rather than the cheap demographic profile we test. On the cautionary side, LLM “survey responses” latch onto spurious option ordering rather than meaning [5]; persona variables explain less than 10% of the variance in subjective

annotation and help only when the persona–outcome correlation is strong [9]; assigning personas can surface biases and *degrade* performance on objective tasks [18]; and persona simulations tend toward *caricature* [2]; and ad-hoc persona *generation* itself injects systematic bias, with election and survey forecasts deviating from real outcomes [12]. Our work contributes a direct *sim-to-real validity* test against measured real-world behaviour (not survey distributions), and—unlike prior work focused on accuracy or annotation—isolates the marginal effect of persona conditioning with a no-persona baseline on a copy-ranking task.

The Upworthy archive and headline performance. The Upworthy Research Archive [13] has supported large-scale analyses of what makes headlines succeed; for example, negativity in headlines increases consumption at scale [15]. These works model *aggregate* drivers of clicks; we instead use the archive as a held-out benchmark to test whether an external persona-based simulator can *predict* the within-test winner. Most directly, concurrent work on the same archive [17] reports that pure-LLM headline prediction (prompting, embeddings, fine-tuning) only marginally beats random; our results refine this picture by showing that *after* filtering to A/B tests with a reliable winner, a no-persona LLM ranker is in fact a medium-strength predictor—and that adding personas erases this. The finding that personas can introduce harmful bias echoes work on LLM simulations producing *caricatures* [2].

Construct validity and effect sizes. We frame the intent-vs-engagement mismatch using the classical construct-validity framework [4] and pre-register effect-size thresholds following Cohen [3].

3 Sim-to-Real Validity Framework

Setup. Figure 1 summarises the protocol. A *package* is an A/B test containing two or more copy *variants* shown to the same real audience; each variant v has an observed click-through rate $\text{CTR}(v)$ and a simulator-predicted score $\text{pred_score}(v)$ (Section 4).

Because variants within a package share an audience, the package defines a controlled comparison.

Why ranking, not absolute CTR. We evaluate *within-package ranking* rather than absolute CTR for two reasons. First, CTR levels vary across packages for reasons unrelated to copy (topic, timing, audience),

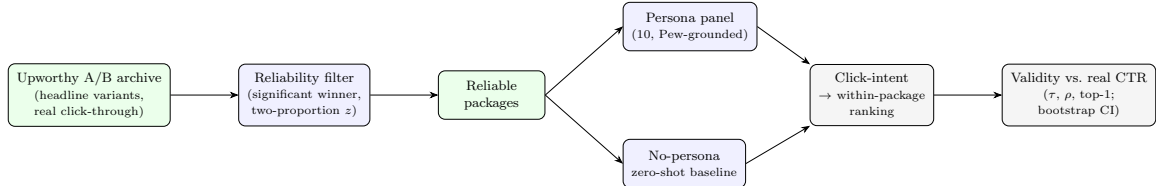


Figure 1: The sim-to-real validity protocol. Real A/B tests are filtered to those with a statistically reliable winner; the persona panel and a no-persona baseline each score the variants, which are ranked within each package and compared to the real click-through ordering.

so only the within-package contrast is attributable to the copy. Second, the simulator’s score and the benchmark outcome are different constructs (intent vs. engagement; Section 7), so a ranking criterion is robust to scale and calibration differences. Concretely, a useful copy simulator need not predict the CTR number—it needs to order candidate variants the way the real audience does.

Metrics. For each package we compare the CTR ordering and the `pred_score` ordering with: top-1 winner accuracy (did the predicted best variant match the real best), Kendall’s τ [10], and Spearman’s ρ [16]. Package-level values are aggregated across packages with bootstrap confidence intervals (Section 5).

4 Method Under Test

We evaluate persona-based copy simulation as a *method*, instantiated as a controlled research harness rather than any specific deployed product: given a piece of copy and an audience, the harness instantiates a panel of synthetic personas and elicits each persona’s reaction, then aggregates these into a ranking. We make the instantiation fully explicit (personas, prompt, and aggregation below) so the result is attributable to the technique, not to an opaque product configuration. The harness and all its inputs are released in the replication package.

Persona panel. The panel is a fixed, versioned set of $P = 10$ personas (Appendix A). To avoid authoring bias, the personas are authored from public demographics rather than reused from any product; they are constructed to represent the real Upworthy-era audience—U.S. social-media news consumers circa 2013–2015—using the demographic structure reported by Pew Research [8] (age, gender, and platform mix, e.g. Facebook news consumers skewing female and Twitter news consumers skewing younger). Each persona is a short natural-language profile (age band,

gender, primary platform, news-engagement level, and a behavioural description).

Elicitation and the predicted score. For each headline variant and each persona we query an instruction-tuned LLM (`gemini-3.1-flash-lite`; selected by the model comparison in Section 5), conditioning on the persona profile and asking, in the persona’s voice, how likely they are to click the headline. The model returns a structured `click_intent` value in $[0, 1]$.

Crucially, the **predicted score** we evaluate is this *click_intent*, aggregated by the **arithmetic mean** over the panel (and over Monte Carlo draws):

$$\text{pred_score}(v) = \frac{1}{PD} \sum_{i=1}^P \sum_{d=1}^D \text{click_intent}_{i,d}(v).$$

We use the simple mean for interpretability; alternative aggregations (median, rank fusion) are a sensitivity dimension we leave to future work (Section 7). We elicit click-intent directly because the benchmark outcome is click-through (engagement); the construct alignment and its limits are analysed in Section 7. Within a package, variants are ranked by `pred_score`; the no-persona baseline (Section 5) uses the identical scoring and aggregation with a single generic reader and no demographic conditioning.

Sampling and reproducibility. Generation uses temperature 0.8 (the simulator’s robust-sampling setting) with a fixed seed forwarded to the model; each Monte Carlo draw uses a distinct seed, and every prediction record stores the model id, seed, temperature, draw count, and panel size for provenance. The reported primary estimate averages $D=3$ draws per persona to control seed-level sampling noise (Section 5).

5 Experimental Setup

Corpus. We use the exploratory split of the Upworthy Research Archive [13]. Rows are grouped

into packages (A/B tests) by test id; within a package we aggregate impressions and clicks per *distinct headline*, since variants that differ only by image cannot be distinguished by a text-only simulator. We keep packages with ≥ 2 distinct headlines and $\geq 3,000$ impressions per headline-variant, the latter for click-through reliability in online controlled experiments [11]. This yields 6978 eligible variants across 1695 A/B tests; the corpus is locked with a content hash and never re-hosted.

Ground-truth reliability filtering. Within-package click-through differences are often tiny and statistically indistinguishable from noise; validating a predictor against a noisy label is ill-posed. We therefore mark a package as having a *reliable winner* when its top-CTR headline significantly beats the runner-up by a one-sided two-proportion z -test ($p < 0.05$). The **primary analysis** uses the reliable-winner packages ($n = 399$); a **sensitivity analysis** reports all predicted packages. This filters on ground-truth reliability, *not* on effect direction, so it does not bias the validity estimate.

Metrics. Ranking agreement is measured *within* each package by top-1 winner accuracy, Kendall’s τ [10], and Spearman’s ρ [16], aggregated across packages with 95% percentile bootstrap confidence intervals (10,000 resamples over packages, seed 42) [6]. The top-1 baseline is the expected accuracy of random guessing given each package’s variant count. We additionally report a decision-relevant metric—the *fraction of achievable CTR lift captured* by the top-1 pick (0 = picks the worst variant, 1 = the real best)—since a copy tool is ultimately judged by the click-through it secures, not only by rank correlation.

Replication and robustness. Beyond the primary (exploratory) split we replicate on two further independent Upworthy splits (holdout, confirmatory) and on two further platforms/domains: MIND news (2019, grouped by subcategory) and Reddit title reposts (2008–12, same image and subreddit, different titles). The same reliable-winner filter (one-sided two-proportion z -test on the per-variant outcome rate) is applied to every dataset—click-through for Upworthy and MIND, upvote ratio for Reddit. We also probe prompt sensitivity with a third-person persona framing and a reworded baseline.

Power and effect-size thresholds. Following Cohen [3], we pre-register $\tau, \rho \geq 0.10$ (with a CI excluding zero) as the bar for a meaningful effect. A power

analysis ($\alpha=0.05$, power= 0.80) gives the required package counts: $n=347$ to detect $\rho=0.15$ and $n=663$ to detect the observed top-1 margin. Using all eligible packages supplies $n = 399$ reliable packages, which powers the rank-correlation claim around $\rho \approx 0.15$; we therefore report top-1 accuracy as an under-powered secondary outcome rather than over-claiming.

No-persona baseline. To isolate the marginal value of persona conditioning, we run a baseline that removes the demographic disaggregation: a single generic “typical U.S. social-media news reader in 2014” prompt with no demographic profile—equivalently, one *aggregate* (population-level) persona—scored and aggregated identically to the panel. For a fair comparison it receives the same per-variant query budget as the panel ($P \times D = 30$ samples per variant, here as 30 draws of the single prompt).

Model selection. A pilot compared `gemin-3.1-flash-lite`, `gemin-2.5-flash`, and `gemin-3.5-flash` on a common subset; their validity confidence intervals were mutually overlapping (statistically indistinguishable), so we select the cheapest and fastest, `gemin-3.1-flash-lite`, for the full study.

Reproducibility and replication. All randomness is seeded (corpus sampling, bootstrap, and LLM generation; seeds in Appendix A). The primary estimate averages $D=3$ Monte Carlo draws (distinct seeds) per persona, because a seed-stability check showed single-seed rank estimates vary by ≈ 0.025 in τ —non-trivial relative to the effect size; the sensitivity-only (non-reliable) packages use a single draw. Predictions are written incrementally to a per-observation checkpoint, making runs crash-safe and resumable.

6 Results

Ground-truth reliability is the binding constraint. Of the 1695 eligible packages, only 399 (23.5%) have a statistically reliable within-package winner (Figure 2).

In the remainder the real click-through difference is indistinguishable from noise, so no predictor can recover the ordering by construction. The primary analysis is therefore restricted to the 399 reliable packages.

Personas vs. a no-persona baseline (RQ1). Table 1 reports within-package ranking agreement (Section 3) on the reliable packages for both methods.

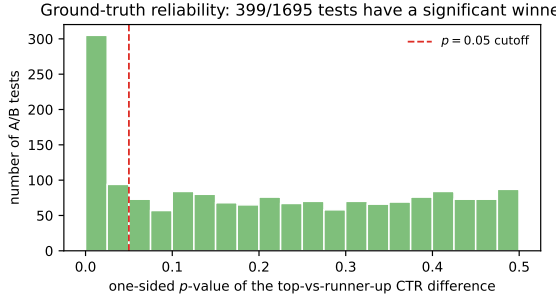


Figure 2: Distribution of the one-sided p -value for the top-vs-runner-up CTR difference across the eligible A/B tests. Only the tests to the left of the dashed cutoff have a statistically reliable winner; the rest cannot be predicted by construction.

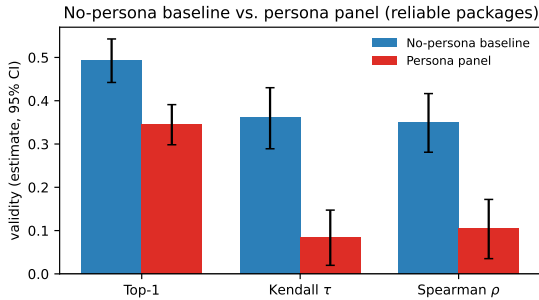


Figure 3: Headline result: the no-persona baseline outperforms the persona panel on every metric, with non-overlapping 95% bootstrap CIs, on the reliable-winner packages.

The headline result is that *persona conditioning hurts*. The no-persona zero-shot baseline ranks variants well—Kendall $\tau = 0.361$ (95% CI [0.289, 0.430]), a medium effect by Cohen’s conventions [3], with top-1 accuracy 49.2% (95% CI [44.2%, 54.3%]) against a random baseline of 30.4%. The ten-persona panel is far weaker: $\tau = 0.084$ (95% CI [0.020, 0.147]) and top-1 34.6%, with confidence intervals that do not overlap the baseline’s on any metric (Figure 3). Persona conditioning thus does not merely fail to help—it actively degrades a usable signal that the base model already has.

The persona panel in isolation. Considered on its own, the panel’s predictive validity is weak but statistically significant (τ and ρ CIs exclude zero), at or below Cohen’s small-effect threshold. Its top-1 CI includes the random baseline, so it does not reliably pick the single best variant; it is, if anything, slightly better at flagging the *worst* variant (37.1%) than the best (34.6%), detecting clearly weak copy more readily

than winners.

Sensitivity analysis. Including all 1695 predicted packages (i.e. without the reliability filter) the effect is weaker but remains significant ($\tau = 0.054$, 95% CI [0.025, 0.083]; $\rho = 0.069$), as expected when noisy-label packages are added.

Does validity track signal strength? (RQ2).

Across 1678 packages spanning the full range of ground-truth signal strength, stronger real effects are associated with slightly higher top-1 success (Spearman 0.059, 95% CI [0.011, 0.107]), while the association with the full rank-correlation is null. The simulator captures a small amount of signal that surfaces mainly when the real effect is large.

Replication across datasets and a different domain (RQ3).

The finding replicates. Table 2 repeats the comparison on three independent Upworthy splits and on a different-domain, more-recent dataset (MIND news, 2019). On all three Upworthy A/B splits the no-persona baseline beats the persona panel by a large, consistent margin (Kendall τ gap between 0.28 and 0.32); the powered confirmatory split ($n = 1866$) gives the cleanest contrast ($\tau = 0.393$ baseline vs. 0.070 persona). On MIND—where “packages” compare *different* articles within a topic rather than headline variants of one story—the direction is the same but the gap is small (0.038), as expected when persona-relevant variance is lower. A boundary case is Reddit title reposts (same image, different titles): there up-votes are dominated by posting time and visibility rather than the title, *neither* method predicts the outcome (both $\tau \approx 0$), and the gap vanishes—the baseline’s advantage appears only where the outcome is actually predictable from the copy. We also report a decision-relevant metric, the fraction of achievable CTR lift captured by the top-1 pick: the baseline captures more than the panel in every dataset (Table 2).

Robustness: seed and model (RQ3).

On a 100-package subset re-run under three seeds, top-1 accuracy is highly stable (SD = 0.008) while the rank coefficient varies by SD = 0.025 in τ —non-trivial relative to the effect size, which is why the primary estimate averages multiple draws. The persona-vs-baseline gap is also consistent across model tiers: on all three **gemin**i models the no-persona baseline beats the persona panel (Table 5), so “personas hurt” is not an artefact of a single model. Nor is it a prompt artefact: a third-person persona framing and a reworded

Table 1: Headline result: a no-persona zero-shot ranker vs. the persona panel on the reliable-winner packages ($n = 399$). The no-persona baseline is substantially better on every metric (non-overlapping CIs). Random top-1 baseline 30.4%. 95% bootstrap CIs.

Method	Top-1 [95% CI]	Kendall τ [95% CI]	Spearman ρ [95% CI]
No-persona baseline	49.2% [44.2, 54.3]	0.361 [0.289, 0.430]	0.350 [0.281, 0.417]
Persona panel	34.6% [29.8, 39.1]	0.084 [0.020, 0.147]	0.104 [0.035, 0.172]

Table 2: Replication across three independent Upworthy splits and two further platforms/domains (MIND news, 2019; Reddit title reposts, 2008–12). The no-persona baseline beats or matches the persona panel everywhere; the advantage is large on controlled copy A/B tests, small on cross-article news, and vanishes on Reddit where upvotes are not title-predictable (both $\tau \approx 0$). Lift = fraction of achievable CTR lift captured by the top-1 pick (persona/baseline). Kendall τ on significant-winner packages. For cross-dataset comparability all rows use a single draw per persona; the exploratory split’s multi-draw primary estimate is in Table 1.

Dataset	n	Persona τ	Baseline τ	Gap	Lift p/b
Upworthy exploratory (2013–15)	399	0.079	0.361	+0.282	0.45/0.58
Upworthy holdout (2013–15)	402	0.100	0.398	+0.298	0.48/0.56
Upworthy confirmatory (2013–15)	1866	0.070	0.393	+0.323	0.44/0.55
MIND news (2019)	74	0.177	0.214	+0.038	0.47/0.51
Reddit titles (2008–12)	600	-0.009	-0.026	-0.017	0.45/0.33

baseline leave both sides essentially unchanged—the persona panel stays weak under either framing and the baseline stays strong—so the gap is not driven by the specific wording of either prompt.

The reported gap is a conservative lower bound.

The baseline is *less* granular than the panel: on 17.8% of packages all its variants receive an identical score and the top-1 pick is a tie, broken arbitrarily. This barely moves the metric (baseline top-1 49.2% under arbitrary tie-breaking vs. 50.0% under fair $1/k$ tie credit), and on the non-tied packages the baseline reaches 52.9%. Because ties drag the baseline’s score down, the persona-vs-baseline gap we report is, if anything, an *under*-estimate of the baseline’s true advantage.

Robustness: aggregation rule (RQ3). The primary estimate pools the panel with an arithmetic mean; to rule out that the result is an artefact of this one choice, we re-aggregate the *same* per-persona observations under six rules (Table 3). None recovers baseline validity—every persona configuration’s τ confidence interval lies entirely below the baseline. Two patterns are informative. First, the panel median lifts τ above the panel mean (Table 3), so the mean is partly dragged by a skewed per-persona score distribution—yet even this best case remains far below the baseline. Second, rank-fusion (Borda) and pair-

wise majority (Condorcet), which discard score *levels* and use only each persona’s *ordering*, land at $\tau \approx 0.08$, no better than the mean; per-persona z -normalisation does not help either. The failure therefore lies in genuine mis-ordering by the personas, not in per-persona scale or offset that a smarter pooling rule could correct.

7 Discussion and Threats to Validity

Why persona conditioning hurts. The base model, asked directly, holds a usable population-level prior on what gets clicked—plausibly because clickability patterns are abundant in its pretraining data. Conditioning on a specific persona (“you are a 68-year-old retiree...”) reframes the task as first-person role-play, which substitutes an idiosyncratic, stereotyped guess for that prior; averaging across a ten-persona panel does not recover it, because each response is biased rather than merely noisy—a caricature effect documented for LLM persona simulations [2], where persona conditioning can surface implicit bias and degrade performance on objective tasks [7]. Seen another way, our no-persona baseline is itself a single *aggregate* persona (a “typical reader”); the finding is then that *disaggregating* the audience into a demographic panel hurts an *aggregate* prediction task—

Table 3: H5 — robustness to the aggregation rule. Six rules applied to the *same* per-persona observations on the reliable-winner packages ($n = 399$), with the no-persona baseline as reference. No rule recovers baseline validity: every persona CI upper bound sits below the baseline. Median best mitigates the panel mean’s outlier sensitivity (τ 0.084 $\rightarrow$ 0.141), while rank-fusion (Borda) and pairwise majority (Condorcet) — which use only persona *orderings* — do not help, locating the failure in genuine mis-ordering rather than per-persona scale. The mean-row top-1 is recomputed here from draw-collapsed per-persona scores and can differ from Table 1’s multi-draw primary estimate by a single package through tie-breaking; the rank coefficient τ is identical. 95% bootstrap CIs.

Aggregation	Kendall τ [95% CI]	Top-1 [95% CI]
No-persona baseline	0.361 [0.289, 0.430]	49.2% [44.2, 54.3]
Panel mean (<i>primary</i>)	0.084 [0.020, 0.147]	34.8% [30.1, 39.3]
Panel median	0.141 [0.076, 0.205]	39.3% [34.6, 44.1]
Trimmed mean (20%)	0.081 [0.017, 0.144]	34.3% [29.6, 38.8]
Per-persona z -mean	0.064 [0.001, 0.128]	33.6% [29.1, 38.3]
Borda (rank fusion)	0.081 [0.018, 0.146]	35.3% [30.8, 40.1]
Condorcet (pairwise)	0.079 [0.014, 0.146]	35.8% [31.3, 40.6]

each disaggregated draw adds role-play bias without adding signal the aggregate query lacks. Empirically this is what we see: the panel’s scores are no less spread within a package than the baseline’s (within-package SD 0.054 vs. 0.054), so personas err by *systematic mis-ordering*, not indecision. In 126 of 398 reliable packages the baseline ranks the true winner first while the panel does not; these are typically curiosity-gap or feel-good headlines (Appendix, Table 6). This is consistent with silicon-sampling working best for *distributional* questions about populations [1] while underperforming for a *point* prediction of aggregate engagement, where the unconditioned prior is the better estimator. It also aligns with independent reports that personas in prompts do not improve—and can degrade—performance [18], and that the persona effect is small and scales with the persona–outcome correlation [9]. The latter predicts our domain-dependence (Section 6): the baseline’s advantage is large on controlled copy A/B tests and shrinks where persona-relevant variance is lower.

Construct validity (RQ3). We elicit a *click-intent* signal matched to the benchmark’s engagement (click-through) outcome, and claim relative ranking rather than absolute CTR [4]. A residual construct gap remains—headlines are editorial, and a *stated* click-intent is not identical to *revealed* clicking—which bounds the achievable validity and plausibly con-

tributes to the small effect size. Tools that instead elicit a purchase- or conversion-intent construct would face an even wider gap to a click outcome; matching the elicited construct to the measured outcome is itself a design lesson.

Statistical-conclusion validity. An initial small-sample pilot overestimated the effect—roughly double the powered estimate—and at the full sample it falls to $\tau = 0.084$, a cautionary instance of small-sample optimism. We pre-registered effect-size thresholds and report bootstrap CIs and seed-stability (SD = 0.025 in τ) to make the weakness of the effect explicit. Top-1 accuracy is under-powered and reported as secondary.

External validity and temporal mismatch. Evidence is from English-language editorial headlines circa 2013–2015, whereas the LLM was trained on a later and broader slice of the web. This temporal gap cuts both ways: the model may have absorbed post-2015 clickbait conventions that inflate the no-persona baseline, or it may have ingested the Upworthy archive itself. We cannot fully rule out either, and flag it as a threat; a contemporaneous benchmark would isolate it. Generalisation to marketing landing-page or ad copy, other languages, and current audiences likewise requires further ground truth (live deployment outcomes or a human panel), which we leave to future work.

Internal validity. Model, temperature, and all seeds are pinned and recorded per prediction; the persona panel is derived from the benchmark audience [8] rather than internal personas to avoid authoring bias (Appendix A). A model comparison found the result stable across three model tiers (Table 4).

Design choices not exhausted. Our estimate fixes one elicitation prompt and one panel size. The aggregation rule, by contrast, is no longer a free parameter: a sweep over six rules—including median, rank-fusion (Borda), and pairwise majority (Condorcet)—leaves the conclusion intact (Table 3, RQ3), and locates the failure in genuine persona mis-ordering rather than the pooling step. Richer psychographic personas, chain-of-thought elicitation, and larger panels could still shift the effect; we report a single well-controlled configuration and leave a sensitivity sweep over those remaining choices to future work. The failure is not specific to the Gemini family: re-running the full protocol on a different family (OpenAI gpt-4.1) reproduces it on all 399 reliable-winner packages—the no-persona baseline again outranks the persona panel

(Kendall τ 0.300 vs. 0.082; top-1 49.1% vs. 35.8%), and the paired per-package baseline—panel gap is significant in both metrics (τ gap +0.214, 95% CI [0.134, 0.296]; top-1 gap +13.2pp, 95% CI [6.6, 19.7]; both exclude 0). Effective persona adoption is therefore not simply an emergent capability the tested tiers lack: a larger frontier model from an unrelated family fails the same way. Whether *any* model configuration makes persona conditioning beneficial for this task remains open. The companion baseline (Section 6) isolates the value of persona conditioning against a no-persona zero-shot ranker.

8 Conclusion

We tested whether persona-based copy simulation predicts how a real audience ranks marketing copy, using the Upworthy A/B-test archive as held-out ground truth, and whether the persona machinery helps at all. Two findings stand out. First, ground-truth reliability is the binding constraint: most A/B tests lack a statistically distinguishable winner, so validity can only be assessed on the reliable subset. Second—counter to the premise of persona simulation—conditioning on personas *degrades* predictive validity: a no-persona zero-shot ranker achieves a medium-sized rank correlation ($\tau = 0.361$) and 49.2% top-1 accuracy, while the demographically grounded ten-persona panel reaches only $\tau = 0.084$ and 34.6%, with non-overlapping confidence intervals. For predicting aggregate engagement, the lesson is to query the model’s population-level prior directly rather than role-play specific personas. This does not rule out personas for goals where heterogeneity matters (segmentation, qualitative objection-finding), but for ranking copy by expected clicks they are, today, worse than not using them. The artifact-first replication package makes every number reproducible, and the protocol—reliability filtering, construct-matched elicitation, and a no-persona baseline—transfers to stronger ground truth and to other copy domains.

Data and Code Availability

The Upworthy Research Archive [13] is publicly available; we redistribute only a locked subset manifest (content hash + sampling parameters), not the source data. The complete simulation harness (personas, prompts, aggregation), all analysis, ingestion, and figure/table-generation scripts, and aggregated result artifacts are released as an artifact-first replication package sufficient to regenerate every reported number. Only the raw per-persona LLM generations (large

volume) are withheld; the harness reproduces them. The replication package is available at <https://github.com/alemaiorano/sim-to-real-validity>.

Ethics and Conflict of Interest

The study uses a public, de-identified archive of A/B-tested headlines; no human subjects were recruited and all personas are synthetic. The author has a commercial interest in copy-simulation tools; this evaluates the general method (a controlled harness), not a specific product, and we mitigate the conflict by pre-registering thresholds, reporting all outcomes (including the null top-1 result and the downward revision from the pilot), and releasing the replication package.

AI Tools Disclosure

- **Language models:** Claude Opus 4.8 (Anthropic, via Claude Code) and the GPT-5 family (OpenAI, via Codex) were used to generate and review code implementations and to refine manuscript text. The method under test is driven by Google Gemini models (**gemini-3.1-flash-lite** as the primary evaluation workhorse, with **gemini-2.5-flash** and **gemini-3.5-flash** as cross-tier robustness comparators) and by OpenAI **gpt-4.1** as the independent cross-family replication model.
- **Web search:** MCP Tavily integration was used to support literature review and fact-checking during manuscript preparation.

All work was carried out under full human oversight and accountability; all experimental results derive from executed code over the public benchmark, and all citations were verified against Crossref/arXiv.

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A Persona Panel, Prompt, and Reproducibility Details

Persona panel and how it was constructed.

The panel is $P=10$ fixed personas. They were authored from aggregate demographics by a deterministic, documented procedure rather than ad hoc, to limit

authorial bias: (i) we took the audience descriptors reported by Pew for U.S. social-media news consumers in the Upworthy era [8]—age band, gender, primary platform (Facebook, Twitter, YouTube, Reddit), and news-engagement level (incidental/moderate/active); (ii) we allocated the ten persona slots to cover this joint space in rough proportion to the reported population (e.g. Facebook as the dominant news gateway, Twitter news consumers skewing younger, Facebook news consumers skewing female); (iii) for each cell we wrote a single behavioural sentence describing *clicking* tendencies, deliberately avoiding brand-specific content; the panel is authored from public demographics and is not reused from any product. The complete, versioned panel ships in the replication package (`data/persona_panel.json`) with the grounding facts recorded inline; the procedure is fully specified there so the panel can be regenerated or extended. We treat persona *psychographics* (beyond demographics) as out of scope and a direction for future work.

Elicitation prompt (verbatim template).

You are simulating a specific person reacting to a news headline on social media in 2014.
 Person: {persona description} (age {band}, {gender}, mainly on {platform}, news engagement: {level}).
 Headline: "{headline}"
 As this exact person, how likely are you to click this headline? Respond ONLY with JSON: {"click_intent": jnumber between 0 and 1_i}.

Predicted score. $\text{pred_score}(v) = \frac{1}{PD} \sum_{i,d} \text{click_intent}_{i,d}(v)$ over P personas and D draws. Variants are ranked by `pred_score` within each package.

Seeds. Corpus sampling seed 42; bootstrap seed 42 (10,000 resamples); LLM generation seed 42+ d for draw d (forwarded to the model’s `generationConfig`). Each prediction record stores model id, seed, temperature (0.8), draw count, and panel size.

Coverage and parse failures. LLM JSON parse failures are rare (< 1% of calls) and retried; a package enters the metrics only when *all* its variants are scored (matching the primary join), so no package is ever ranked from partial coverage. Per-persona draws that fail are simply averaged over the remaining draws.

Ground-truth reliability test. A package has a reliable winner when its top-CTR headline beats the runner-up by a one-sided two-proportion z -test at

$p < 0.05$ [11]. 23.5% of eligible A/B tests pass this filter.

Model comparison (pilot). Table 4 reports the pilot that motivated the model choice: three `gemin` tiers on a common subset, with overlapping validity intervals.

Cross-model robustness. Table 5 shows the persona-vs-baseline gap on all three model tiers; the baseline wins on every one. All runs disable model thinking (`thinkingBudget=0`): the task is a single scalar rating, and `flash-lite` performs no thinking by default, so this makes the configuration explicit and the comparison fair and inexpensive.

Error-analysis examples. Table 6 shows real winning headlines the no-persona baseline ranked first but the panel did not, illustrating the systematic mis-ordering discussed in Section 7.

Full persona panel. Table 7 lists the ten personas; full behavioural descriptions ship in the replication package (`data/persona_panel.json`).

Table 4: Pilot model comparison on a common subset ($n = 100$ packages). The validity intervals overlap across model tiers, so the cheapest model was selected. 95% bootstrap CIs.

Model	Top-1	Kendall τ	Spearman ρ
gemini-2.5-flash	0.390 [0.29, 0.49]	0.116 [-0.01, 0.24]	0.133 [-0.01, 0.27]
gemini-3.1-flash-lite	0.330 [0.24, 0.42]	0.035 [-0.10, 0.17]	0.051 [-0.09, 0.19]
gemini-3.5-flash	0.390 [0.29, 0.49]	0.122 [-0.01, 0.25]	0.139 [-0.00, 0.28]

Table 5: Cross-model robustness ($n = 100$ reliable packages, Kendall τ). The no-persona baseline beats the persona panel on every model tier; persona conditioning hurts regardless of model.

Model	Persona τ	Baseline τ	Gap
gemini-3.1-flash-lite	0.035	0.401	+0.367
gemini-2.5-flash	0.116	0.219	+0.103
gemini-3.5-flash	0.122	0.404	+0.282

Table 6: Examples of real winning headlines that the no-persona baseline ranked first but the persona panel did not (of 126 such cases). These are typically curiosity-gap / feel-good headlines the base model recognises as clickable.

Real winning headline	CTR	#var
Bookmark This For When You're Sad And Instantly Be 100 Times Happier	1.38%	4
10 Amazing International Photos That Will Give You Some Perspective	1.92%	4
If You Saw Someone Bullying A Kid For Being Gay Would You Speak Out...	1.11%	2
If You Watch Hilary Swank Annoy A Whiny Senator, \$1 Will Go To Pun...	1.67%	3
Seven Wonderful People Share Secrets That Are Killing Them. This C...	1.99%	4
A 9-Year-Old Asks McDonald's CEO A Simple Question. She's Not Lovin...	3.23%	4
Did These Words Really Come Out Of Conan's Mouth?	4.45%	3
We Use This Word So Often We Probably Don't Realize We're Saying It...	1.68%	4
Did You Know You Can Go For A Swim In Google Earth?	0.88%	3
Cookie Monster Is In Jail? I'm Actually Really Happy About The Reas...	2.30%	4

Table 7: The full fixed persona panel ($P=10$), grounded in the Upworthy-era audience demographics [8], with the verbatim behavioural descriptions used in the prompt (for replicability). News engagement and political lean are in the replication package (`data/persona_panel.json`).

ID	Demographics	Behavioural description (verbatim)
p01	18-29, F, Facebook	29-year-old woman, scrolls Facebook on her phone during breaks; clicks human-interest and feel-good stories shared by friends; rarely seeks news directly.
p02	18-29, M, Twitter	24-year-old man, heavy Twitter user, follows journalists and culture accounts; clicks surprising or contrarian takes; low tolerance for clickbait he's seen before.
p03	30-49, F, Facebook	38-year-old working mother, checks Facebook in the evening; drawn to headlines about family, health, and local impact; shares uplifting or outrage-inducing stories.
p04	30-49, M, Reddit	41-year-old man, browses Reddit and Facebook; skeptical of sensational headlines, clicks when a headline promises concrete information or a strong narrative payoff.
p05	50-64, F, Facebook	57-year-old woman, mostly on Facebook; clicks emotionally resonant stories and anything about social causes she follows; less drawn to internet-slang headlines.
p06	50-64, M, Facebook	60-year-old man, reads news on Facebook and cable; clicks political and economic headlines, distrusts overtly hyped 'you won't believe' framing.
p07	18-29, F, YouTube	22-year-old college student, video-first, on YouTube and Instagram; clicks visually evocative or identity-relevant headlines; very sensitive to authenticity.
p08	30-49, M, Facebook	45-year-old man, casual Facebook user; clicks curiosity-gap headlines that promise a quick interesting fact, especially maps, lists, and 'what happened next' stories.
p09	65+, F, Facebook	68-year-old retiree, active on Facebook with family; clicks heartwarming, nostalgic, or cause-driven stories; avoids headlines that feel like tricks.
p10	30-49, F, Twitter	34-year-old professional, follows news and advocacy on Twitter; clicks data-driven or justice-oriented angles; fatigued by repetitive viral formats.